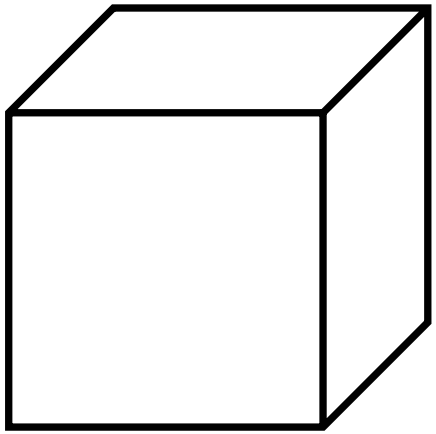
The Promise and Perils of Big Data

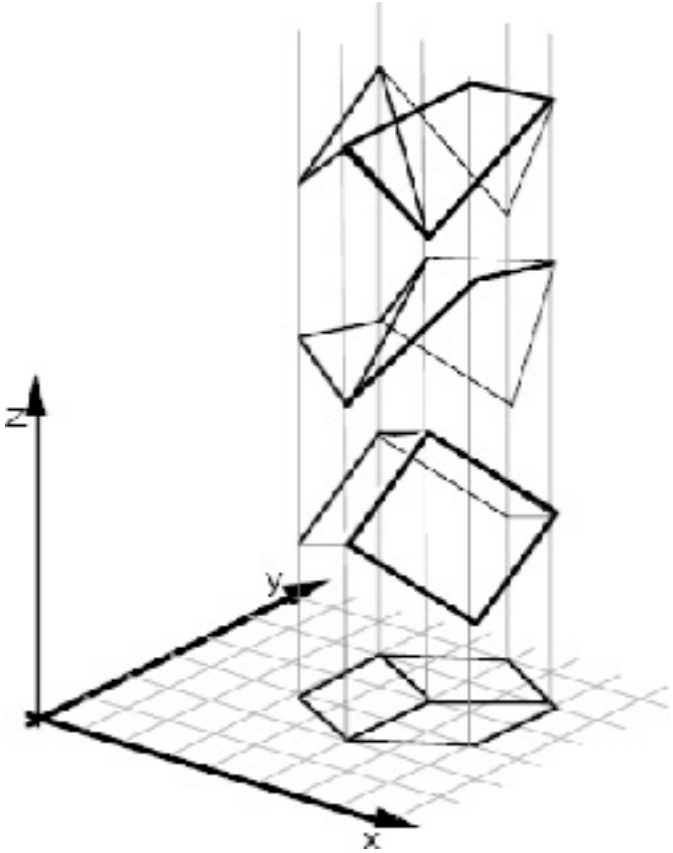


Some Slides from A. Gupta, A. Efros and A. Torralba

Why do we need data?

Most vision problems are ambiguous



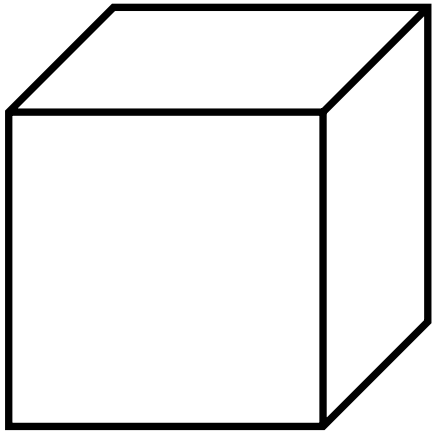


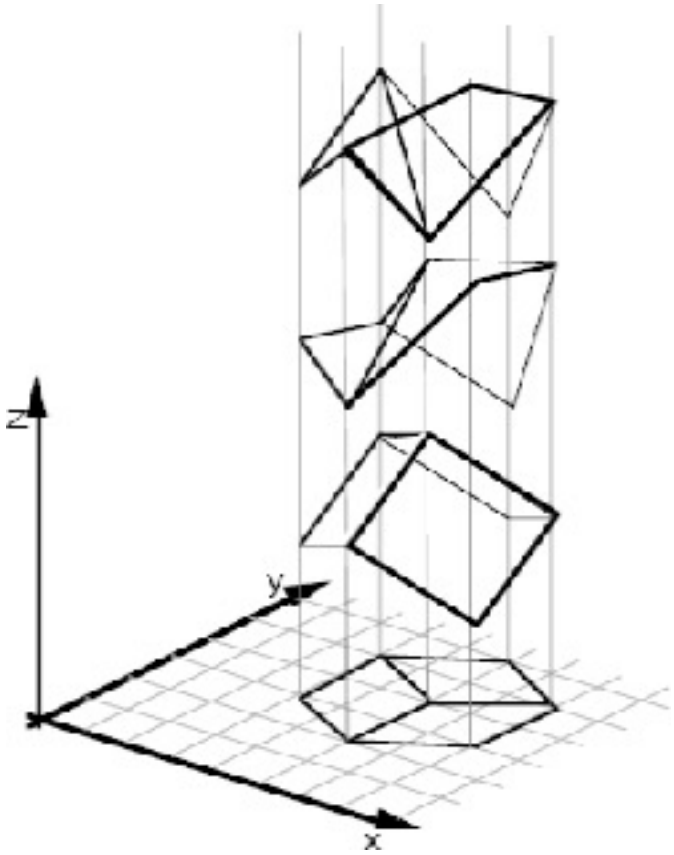
Why do we need data?

Most vision problems are ambiguous



How do we solve these problems?





A cube is *more likely* explanation than some other shapes

How do we solve these problems?

- Magic of data!
- Use data to learn better likelihoods: how things look like
- Use data to learn priors of what is more likely than others

But how much data do we need?

So how much data do humans use?

What's the Capacity of Visual Long Term Memory?

What we know…

Standing (1973)

10,000 images

83% Recognition

… people can remember thousands of images

pigeons: 800-1200
baboons: 3500-5000
(Fagot and Cook, 2006)

Slide by Aude Oliva

What's the Capacity of Visual Long Term Memory?

What we know…

Standing (1973)

10,000 images

83% Recognition

… people can remember thousands of images

High Fidelity Visual Memory is possible (Hollingworth 2004)

What we don't know…

… what people are remembering for each item?



According to Standing

"Basically, my recollection is that we just separated the pictures into distinct thematic categories: e.g. cars, animals, single-person, 2-people, plants, etc.) Only a few slides were selected which fell into each category, and they were visually distinct."



"Gist" Only



Sparse Details



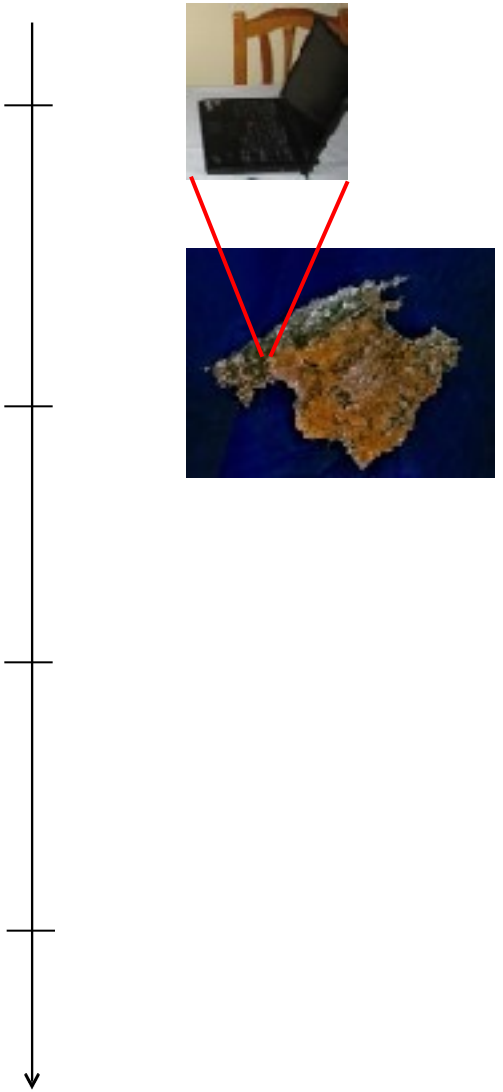
Highly Detailed

Slide by Aude Oliva

Datasets in perspective

Number of images we can memorize: 10^4

Number of images seen during my first 10 years: 10^8
(3 images/second * 60 * 60 * 16 * 365 * 10 = 630720000)



Number of
all samples

How much data does computer vision researchers use?

2004-2006 Caltech 101 and 256



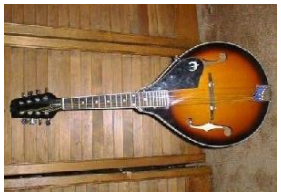






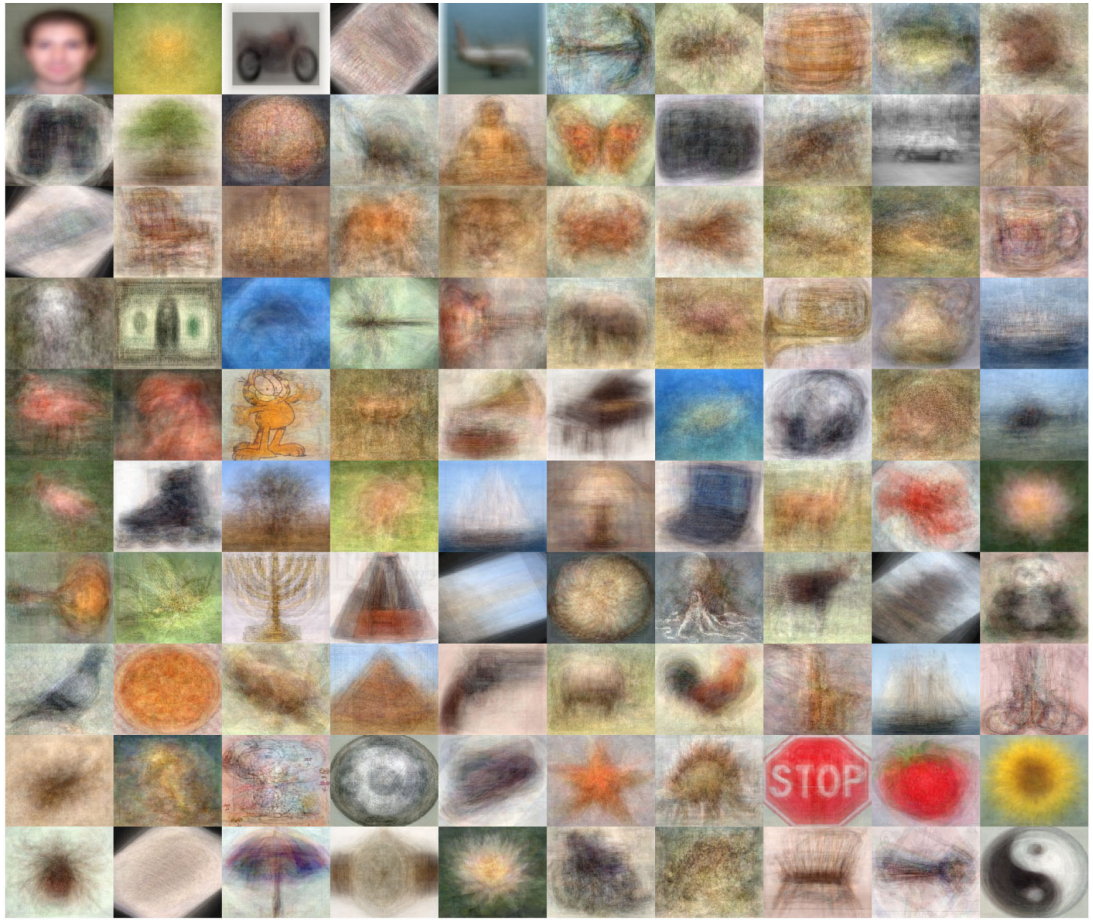







2006 80 Million Tiny Images

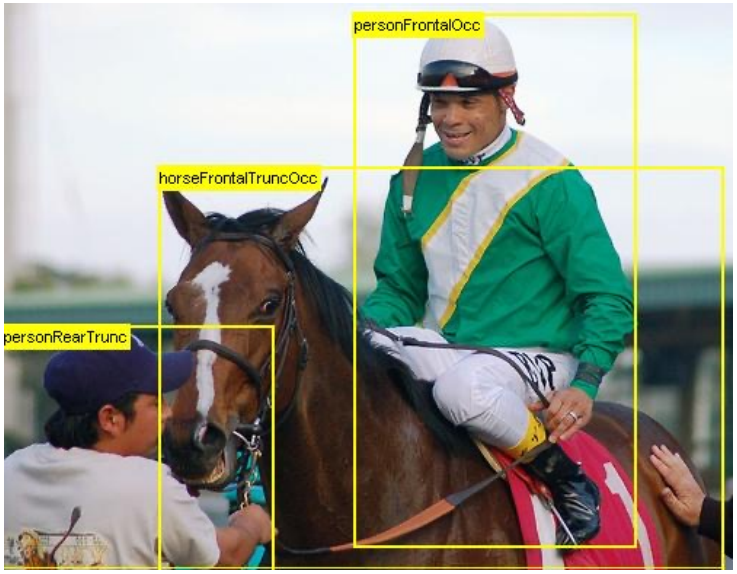
Antonio Torralba, et al. TPAMI 2008



Withdrawn in 2020 due to derogatory categories and offensive images, because of automated data collection.

2007 PASCAL VOC

20 classes





The PASCAL Visual Object Classes (VOC) Challenge, Everingham, Van Gool, Williams, Winn and Zisserman, *IJCV*, 2010

2009 Caltech Pedestrian

1 class, lots of instances.



Pedestrian Detection: An Evaluation of the State of the Art, Dollár, Wojek, Schiele and Perona, *PAMI*, 2012

2009 ImageNet

22K categories, 14M images





ImageNet: A Large-Scale Hierarchical Image Database,
Deng, Dong, Socher, Li, Li and Fei-Fei, *CVPR,* 2009

2010 SUN

908 scene categories

Beer garden



SUN Database: Large-scale Scene Recognition from Abbey to Zoo
Xiao, Hays, Ehinger, Oliva, and Torralba, *CVPR*, 2010.

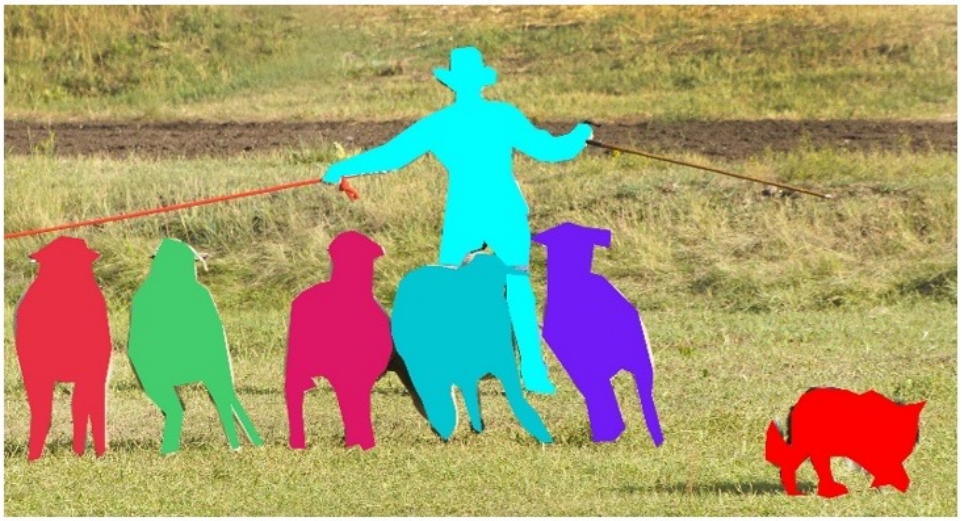
2012 COCO

80 object categories with instance masks

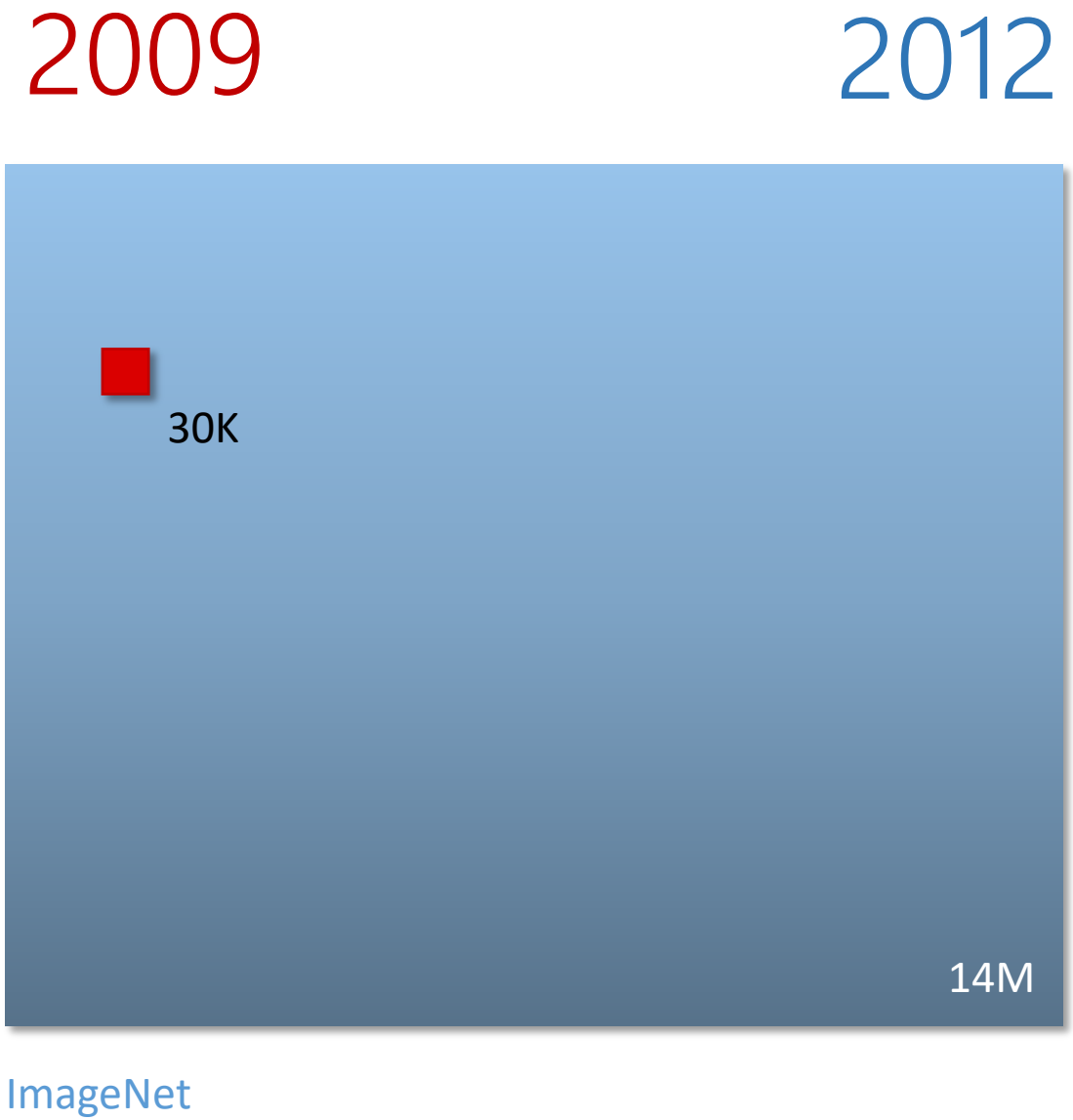
Objects in Context

Instance segmentation

Non-iconic instances



Images in vision datasets



2022 LAION-5B

Images with their captions

Backend url:
https://knn5.laior
Index:
laion_5B

french cat

Clip retrieval works by converting the text query to a CLIP embedding , then using that embedding to query a knn index of clip image embedddings

Display captions ☑
Display full captions ☐
Display similarities ☐
Safe mode ☑
Hide duplicate urls ☑
Hide (near) duplicate images ☑
Search over
image
Search with multilingual clip ☐


french cat


french cat


How to tell if your feline is french. He wears a b...


イケメン猫モデル「トキ・ナンタケット」がかっこいい - NAVER まとめ


Hilarious pics of funny cats! funnycatsgif.com


Hipster cat


網友挑戰「加幾筆畫出最創意貓咪圖片」，笑到岔氣之後我也手


cat in a suit Georgian sells tomatoes




French Bread Cat Loaf Metal Print

Datasets in perspective

Number of images we can memorize: 10^4

Number of images seen during my first 10 years: 10^8
(3 images/second * 60 * 60 * 16 * 365 * 10 = 630720000)

Number of images seen by all humanity: 10^{20}
106,456,367,669 humans[1] * 100 years * 3 images/second * 60 * 60 * 16 * 365
[1 from http://www.prb.org/Articles/2002/HowManyPeopleHaveEverLivedonEarth.aspx](http://www.prb.org/Articles/2002/HowManyPeopleHaveEverLivedonEarth.aspx)

Number of all 32x32 images: 10^{7373}
256 ^{32*32*3} ~ 10^{7373}



When do we need big data?

Two Kinds of Things in the World





Navier–Stokes momentum equation (*convective form*)

$$\rho\frac{D\mathbf{u}}{Dt} = -\nabla \bar{p} + \mu\,\nabla^2\mathbf{u} + \tfrac{1}{3}\mu\,\nabla(\nabla\cdot\mathbf{u}) + \rho\mathbf{g}.$$

Nasdaq 10 Years
Daily Chart

Unreasonable Effectiveness of Data

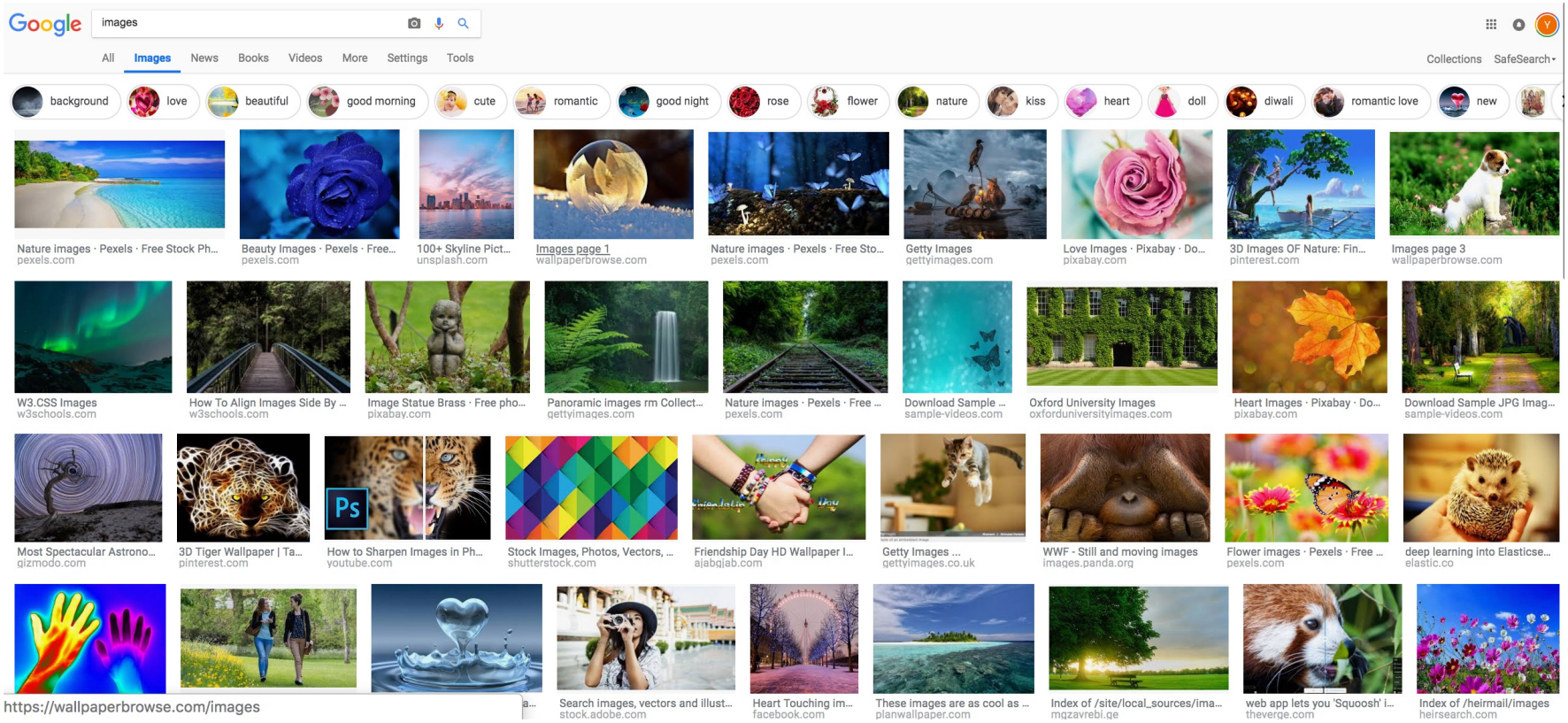
[Halevy, Norvig, Pereira 2009]

- Parts of our world can be explained by elegant mathematics
  - Physics, astronomy, chemistry, etc.

- But much cannot
  - economics, psychology, …
  - visual recognition?

- Enter **The Magic of Data**
  - Great advances in speech, vision, and NLP

Lots of Data Available



Thanks to the Internet

Unreasonable Effectiveness of Data

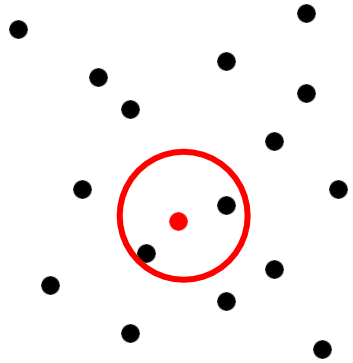
Simple and scalable algorithms (may be dumb) + lot of data

are better than

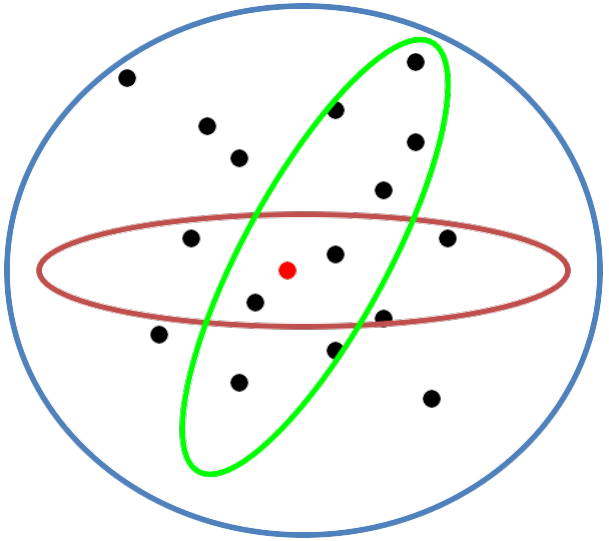
complicated algorithms + small amount of data.

How do we use big data?

Two ways to use Lots of Data

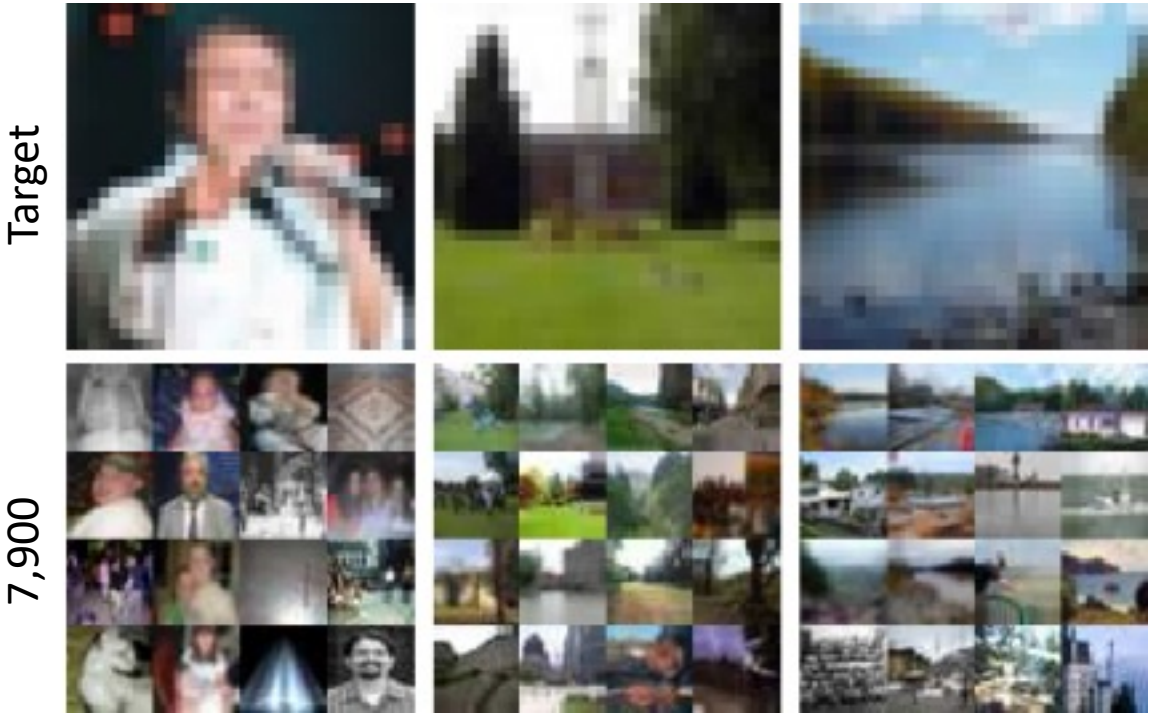


Brute Force Vision: Find that needle in the haystack and disregard the rest (a.k.a. kNN)



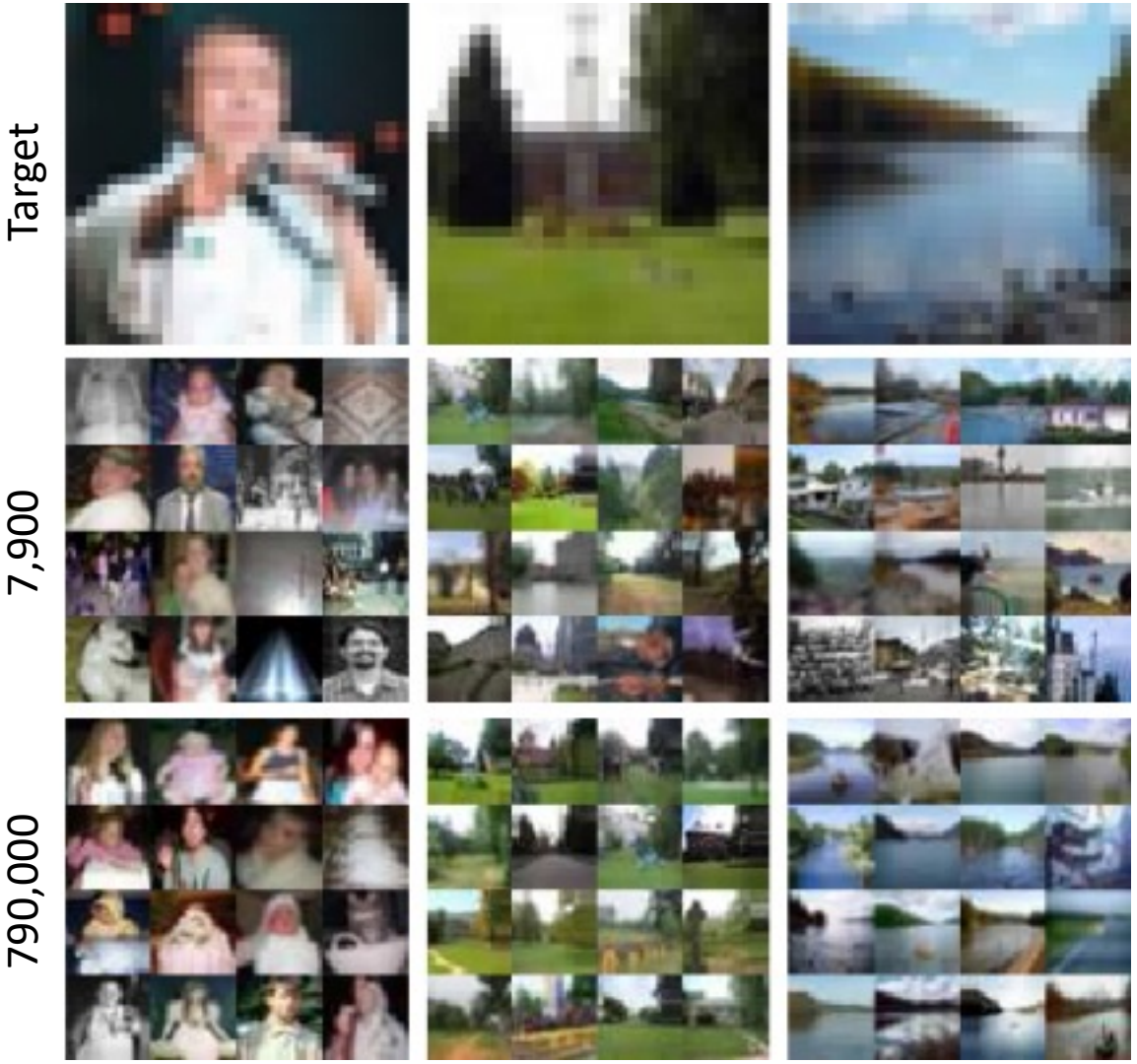
See what different subsets of data think of you (a.k.a. Parametric Models)

Lots Of Images



A. Torralba, R. Fergus, W.T.Freeman. PAMI 2008

Lots Of Images



A. Torralba, R. Fergus, W.T.Freeman. PAMI 2008

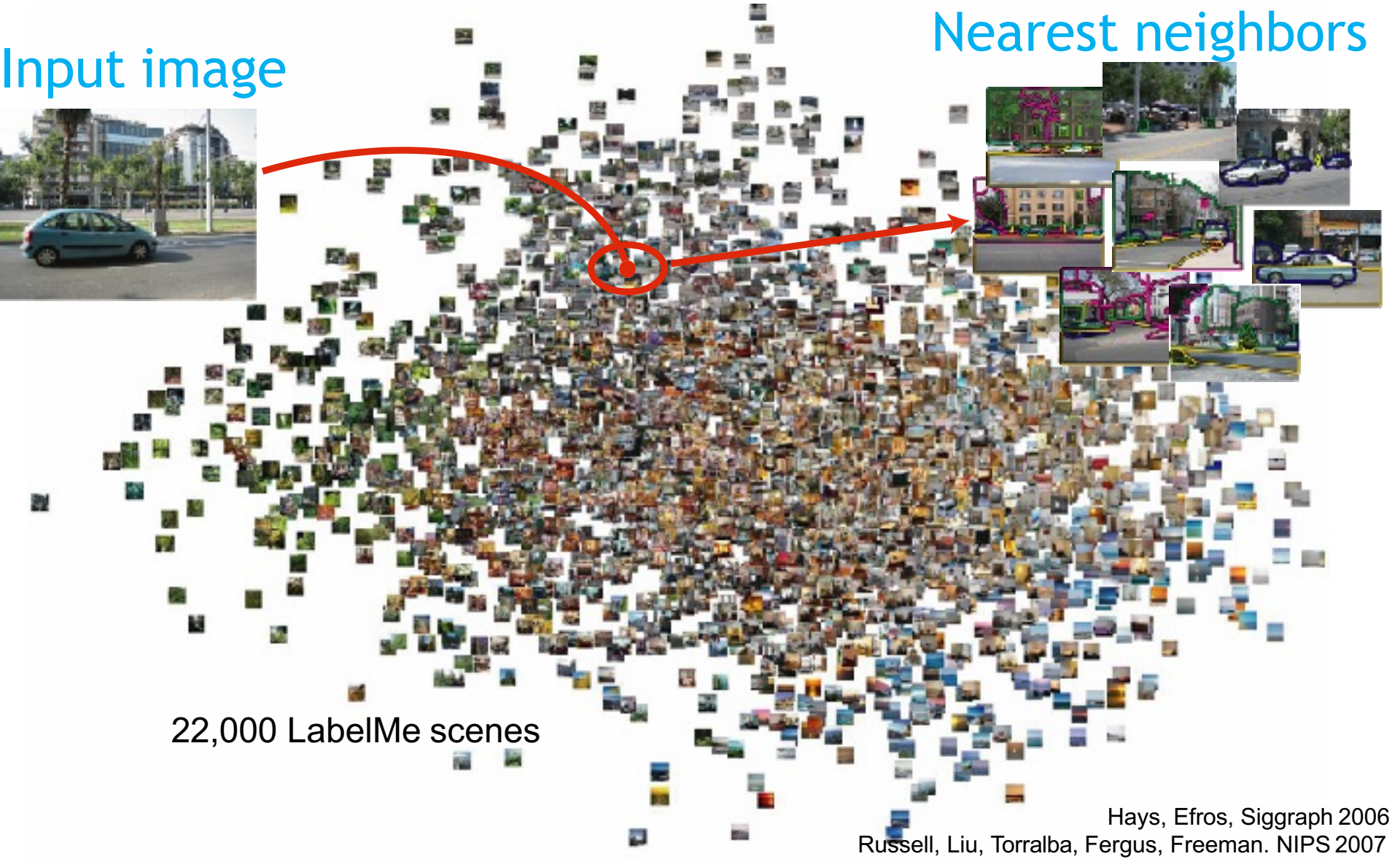
Lots

Of

Images



With a good image similarity and a lot of data…



Hays, Efros, Siggraph 2006
Russell, Liu, Torralba, Fergus, Freeman. NIPS 2007

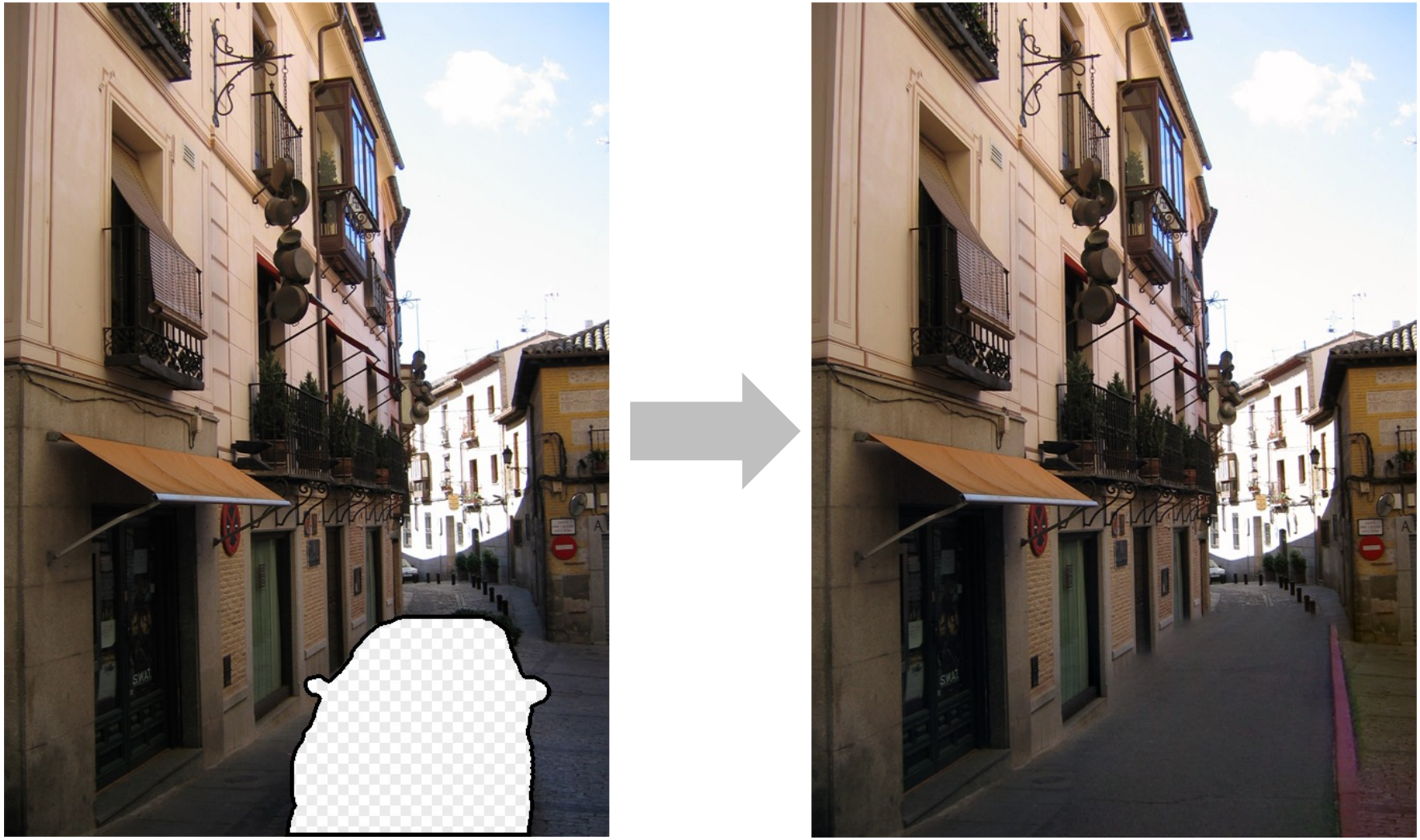
kNN matching is great…

- **Because we live in a (mostly) boring world!**



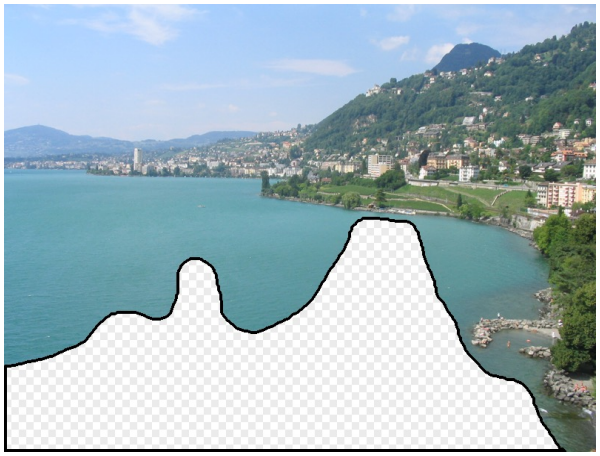


Image Completion

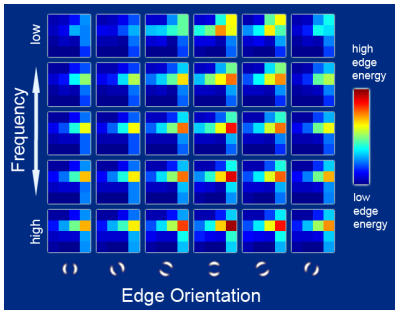


Hays and Efros, SIGGRAPH 2007

Image Completion



Input image

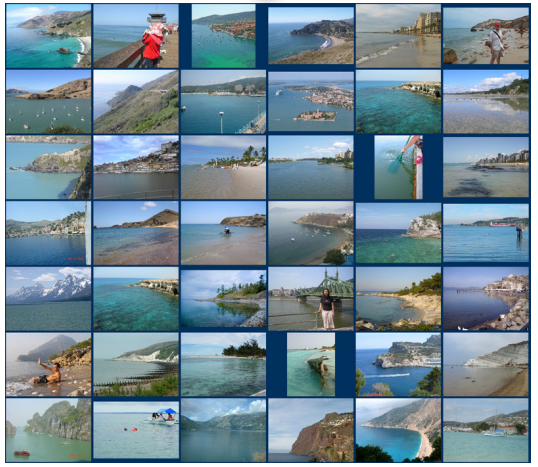




Scene Descriptor



Image Collection



200 matches



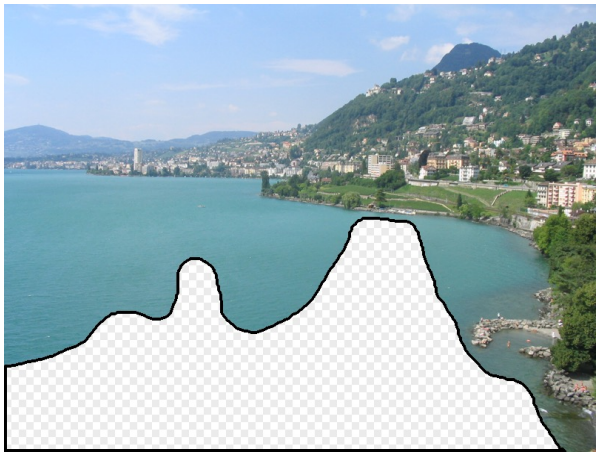
Context matching
+ blending



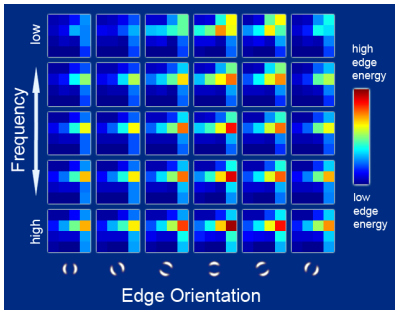
20 completions

Hays and Efros, SIGGRAPH 2007

Image Completion



Input image

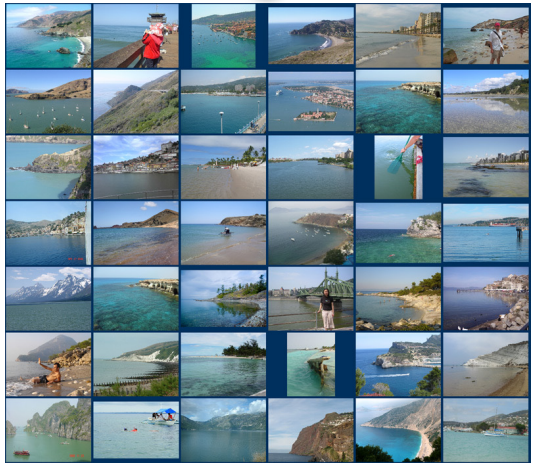




Scene Descriptor



Image Collection



200 matches

Hays and Efros, SIGGRAPH 2007



Context matching
+ blending



Automatic Colorization Result

Grayscale input



Colorization of input using average



A. Torralba, R. Fergus, W.T.Freeman. 2008

While many scenes are boring…











Slide by Antonio Torralba

Some scenes are unique



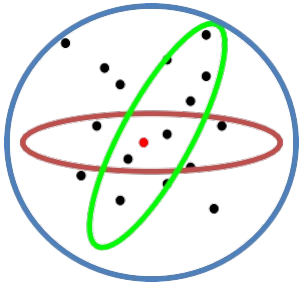




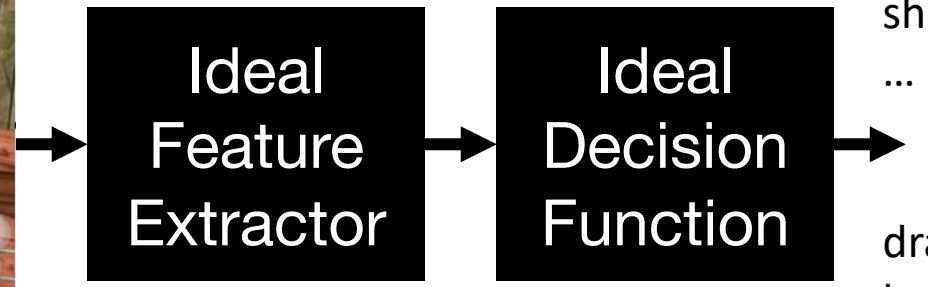
Slide by Antonio Torralba

Generalization to rare scenes

- Move away from kNNs …
- Use data to make connections







window, top-left
clock, top-middle
shelf, left
…

drawing, middle
hat, bottom right

Parametric models



Ideal Feature Extractor → Ideal Decision Function →

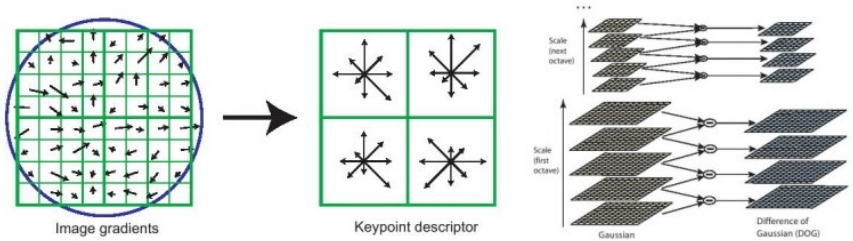
window, top-left
clock, top-middle
shelf, left
…

drawing, middle
hat, bottom right

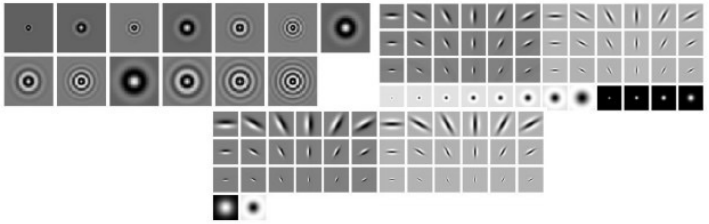
Q: What types of features shall we consider?

Q: What types of decision functions shall we consider?

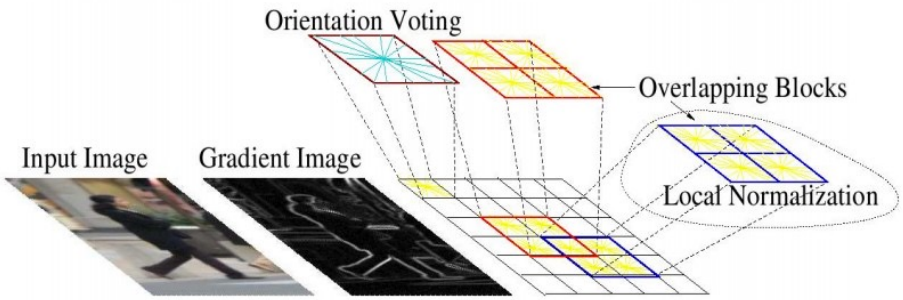
Computer vision features



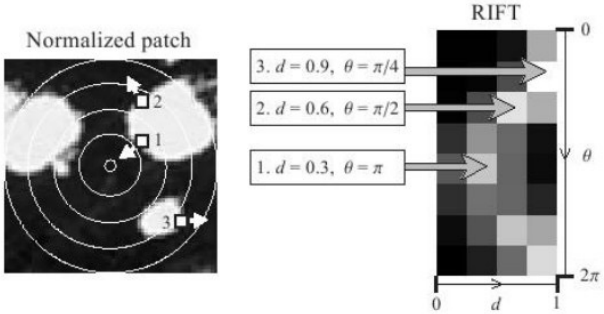
SIFT



Textons

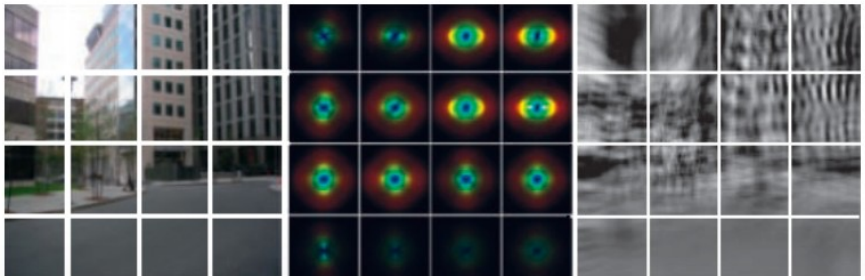


HoG



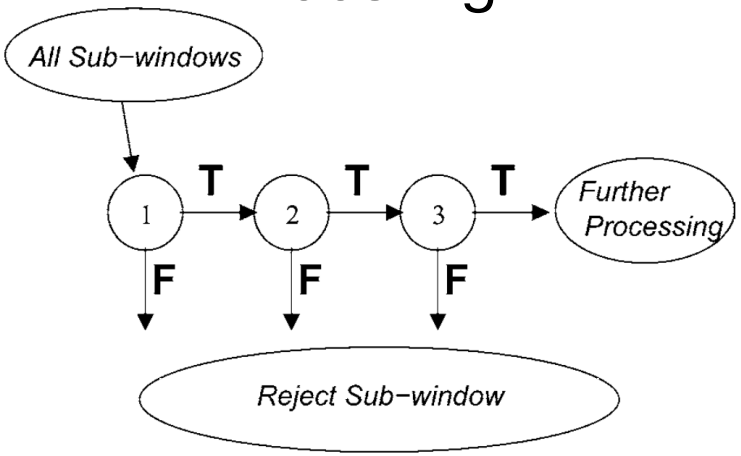
RIFT

GIST

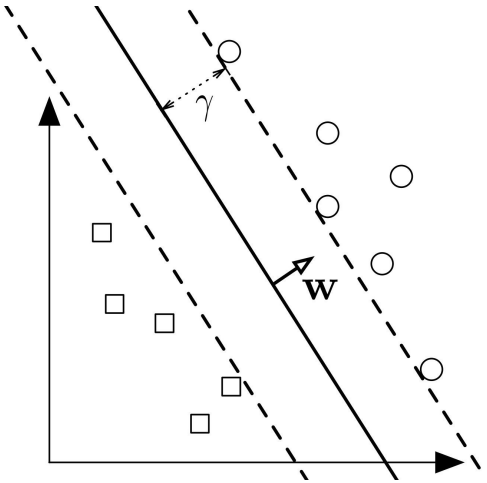


Computer vision decision functions

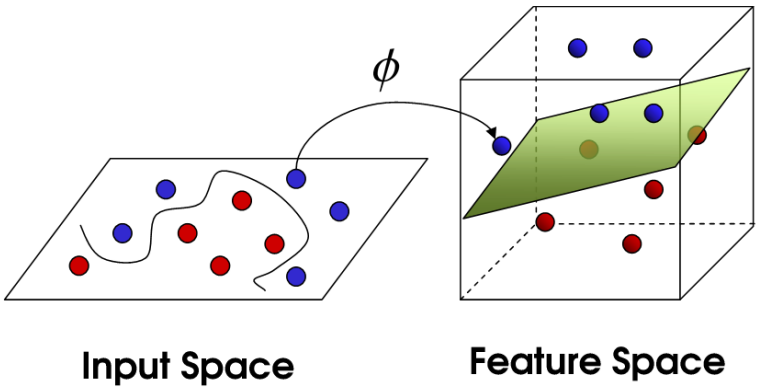
Boosting



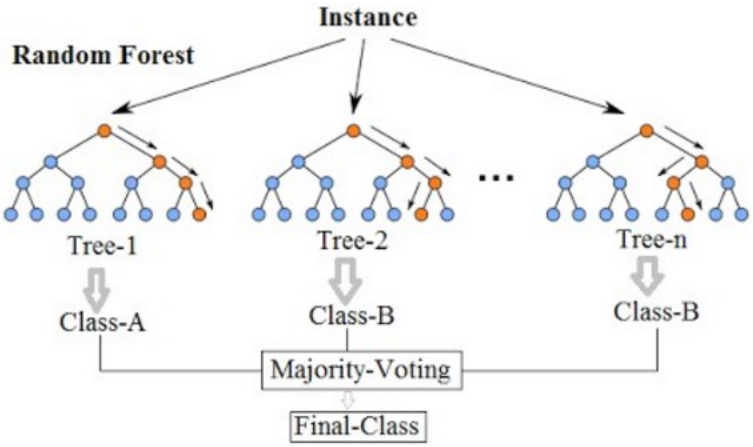
Linear SVM



Kernel Machines



Random Forest



Can we learn everything in one shot?

Representation learning & deep learning

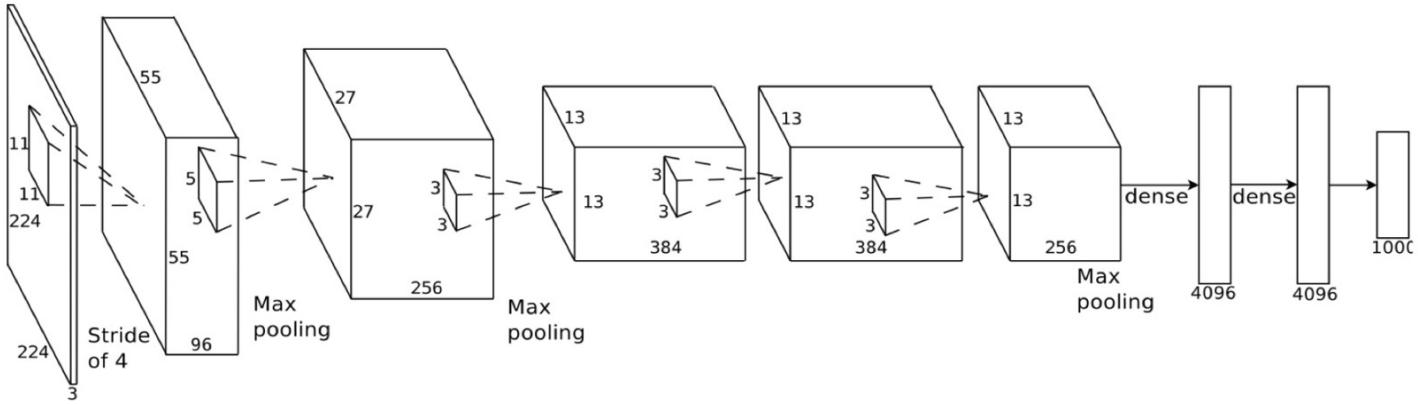
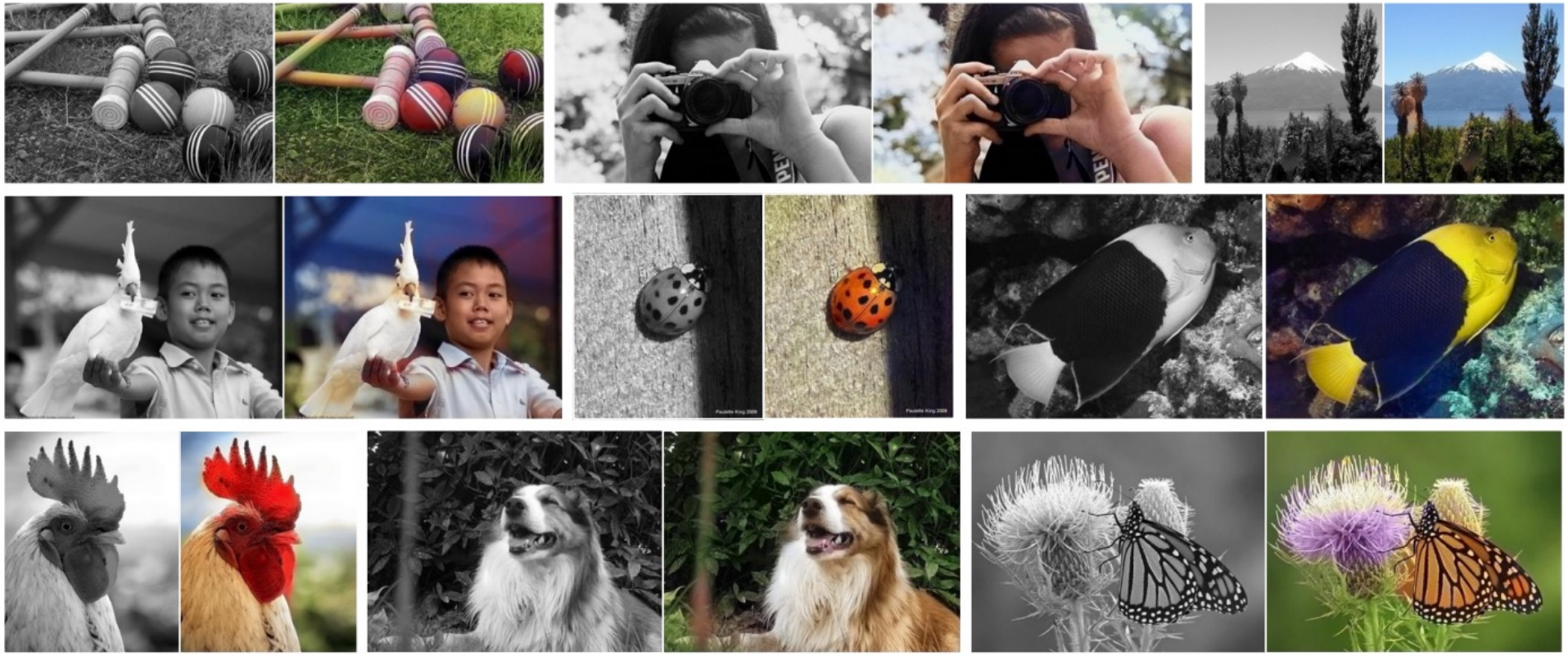


Image completion using deep models



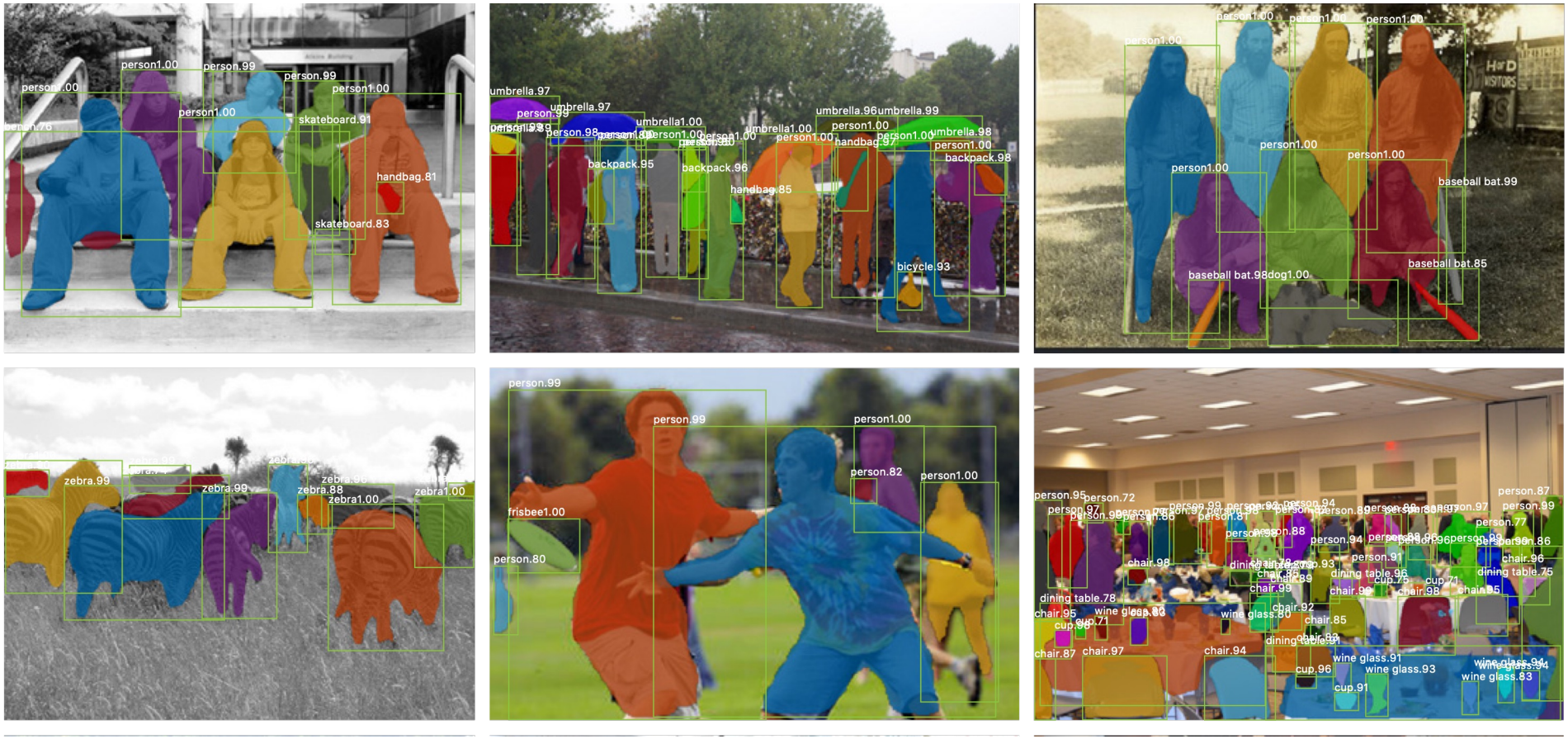
Iizuka, Simo-Serra, Ishikawa, SIGGRAPH 2017

Image colorization using deep models



Zhang, Isola, Efros, ECCV 2016

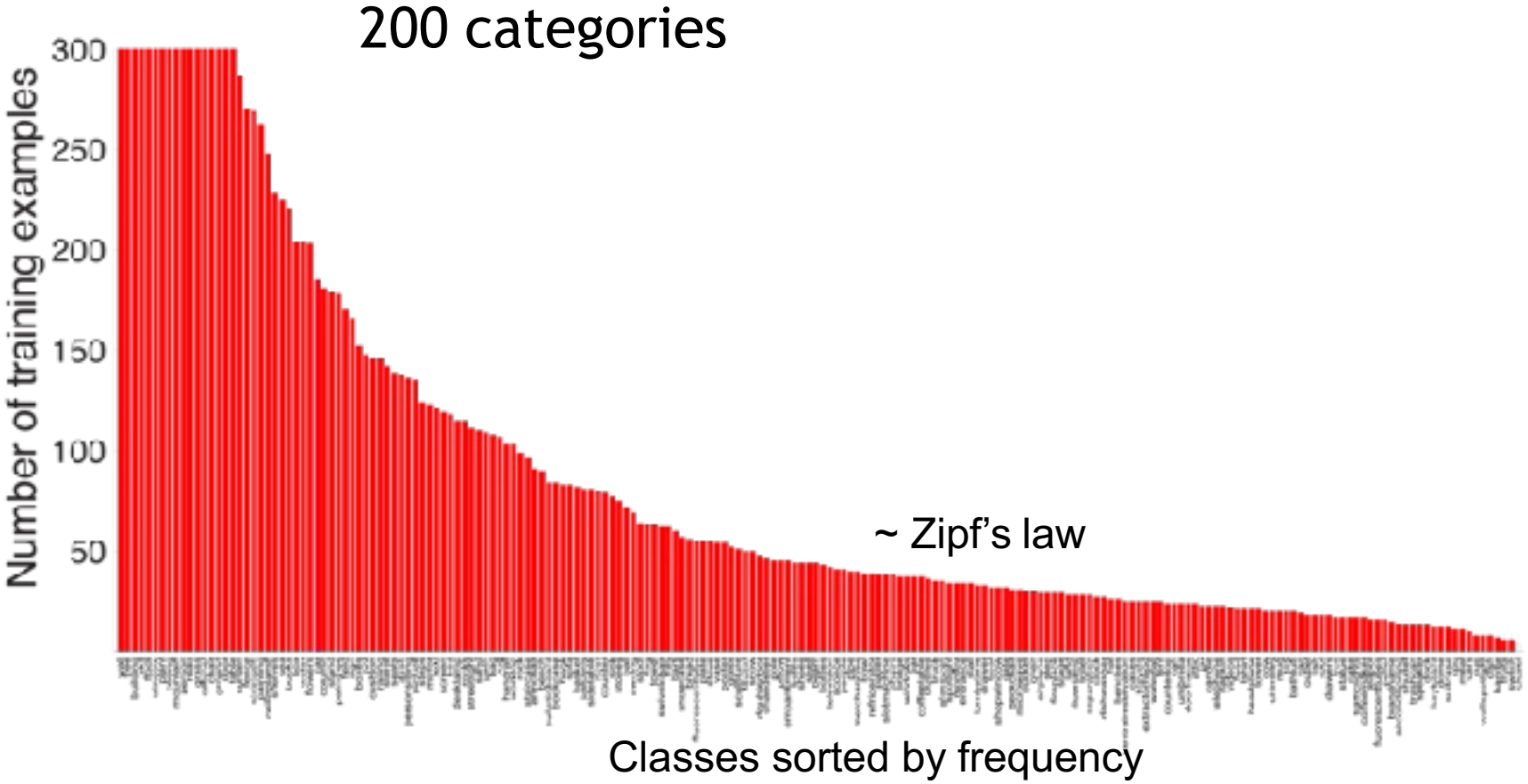
Object detection using deep models



"Mask RCNN", He, Gkioxari, Dollár, Girshick, *ICCV,* 2017

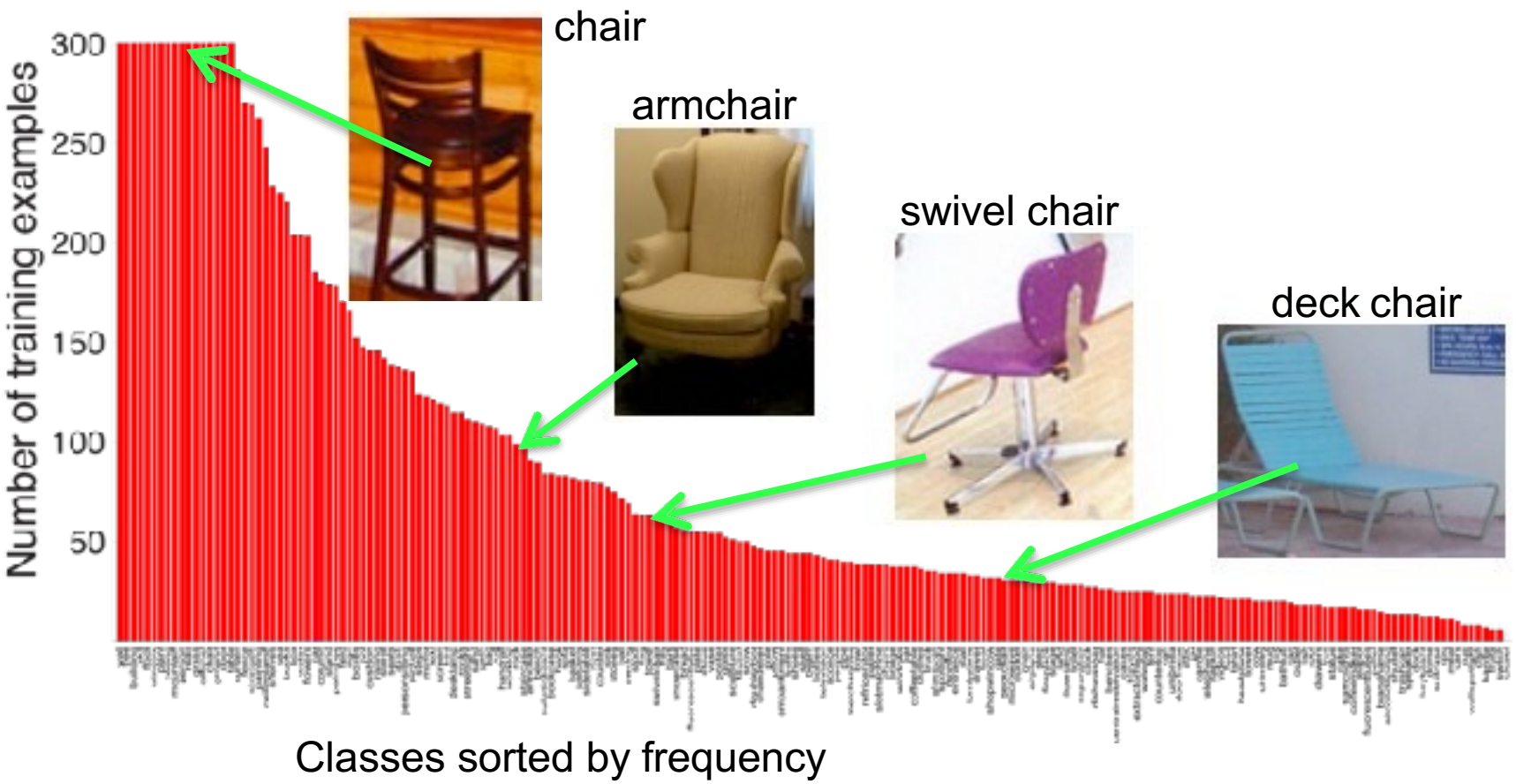
The Bias of Data

SUN dataset



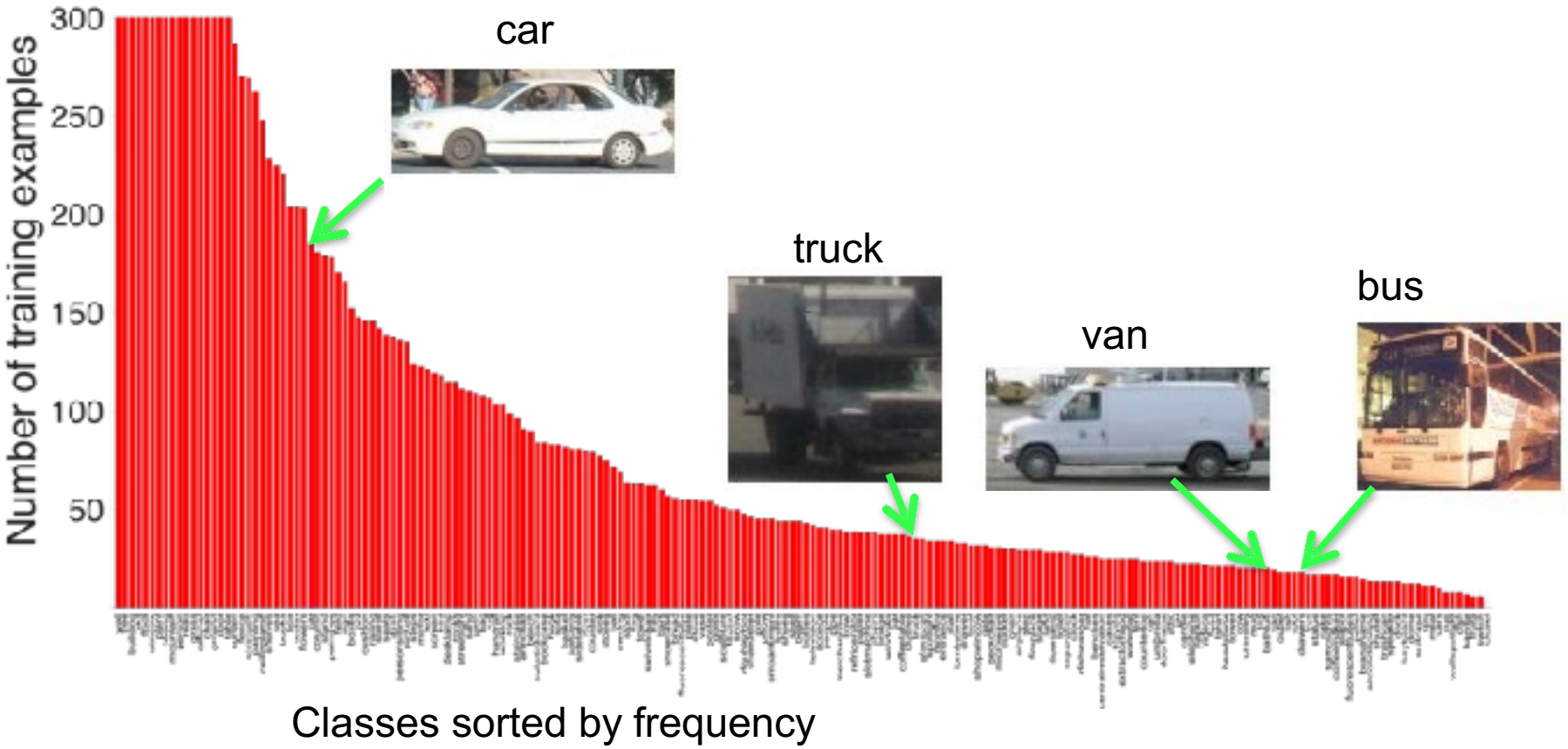
The first 9 objects account for 50% of all training examples
17 classes with more than 300 examples
109 classes with less than 50 examples

Rare objects are similar to frequent objects



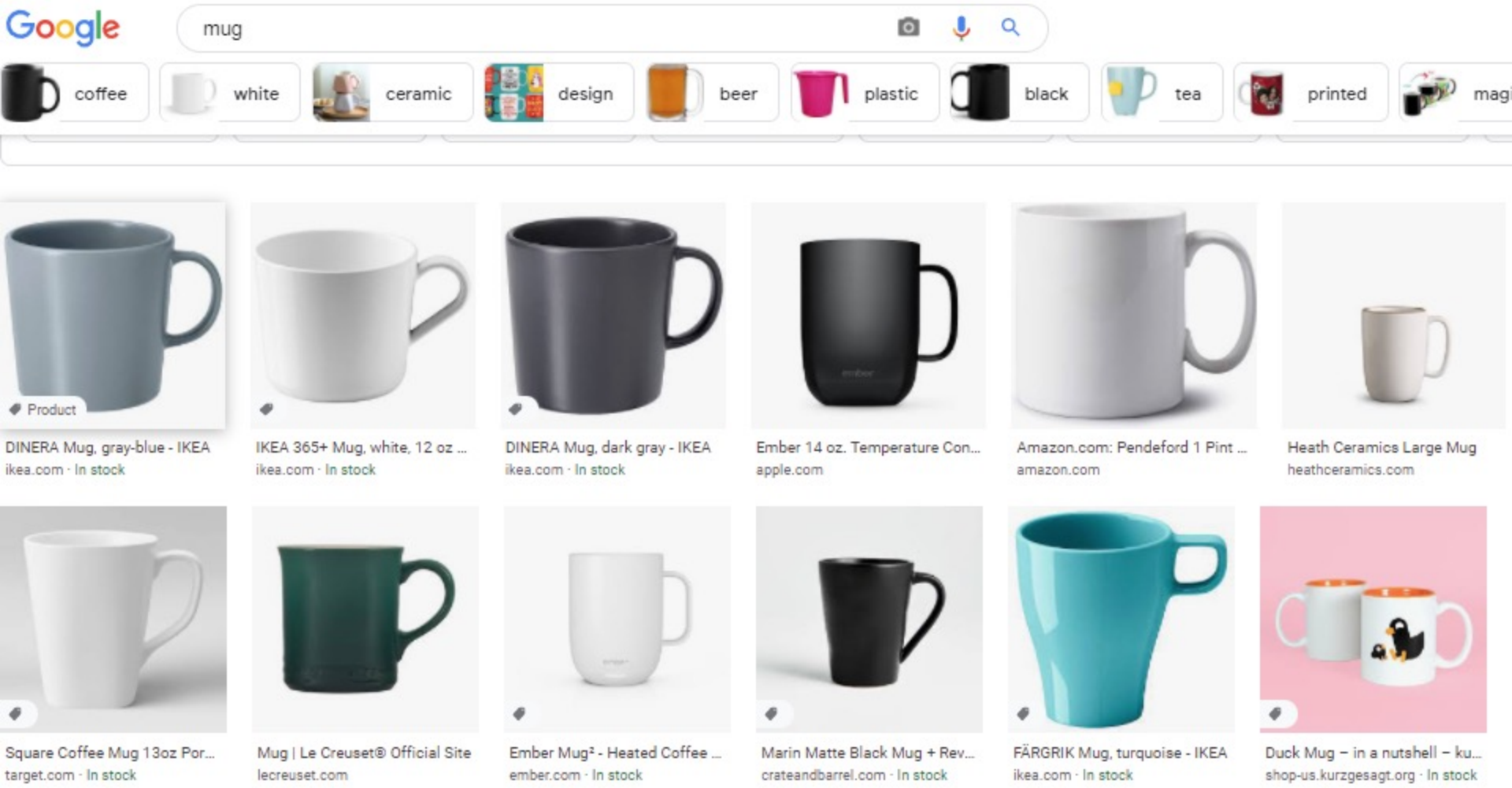
Salakhutdinov, Torralba, and Tenenbaum, CVPR, 2011

Rare objects are similar to frequent objects

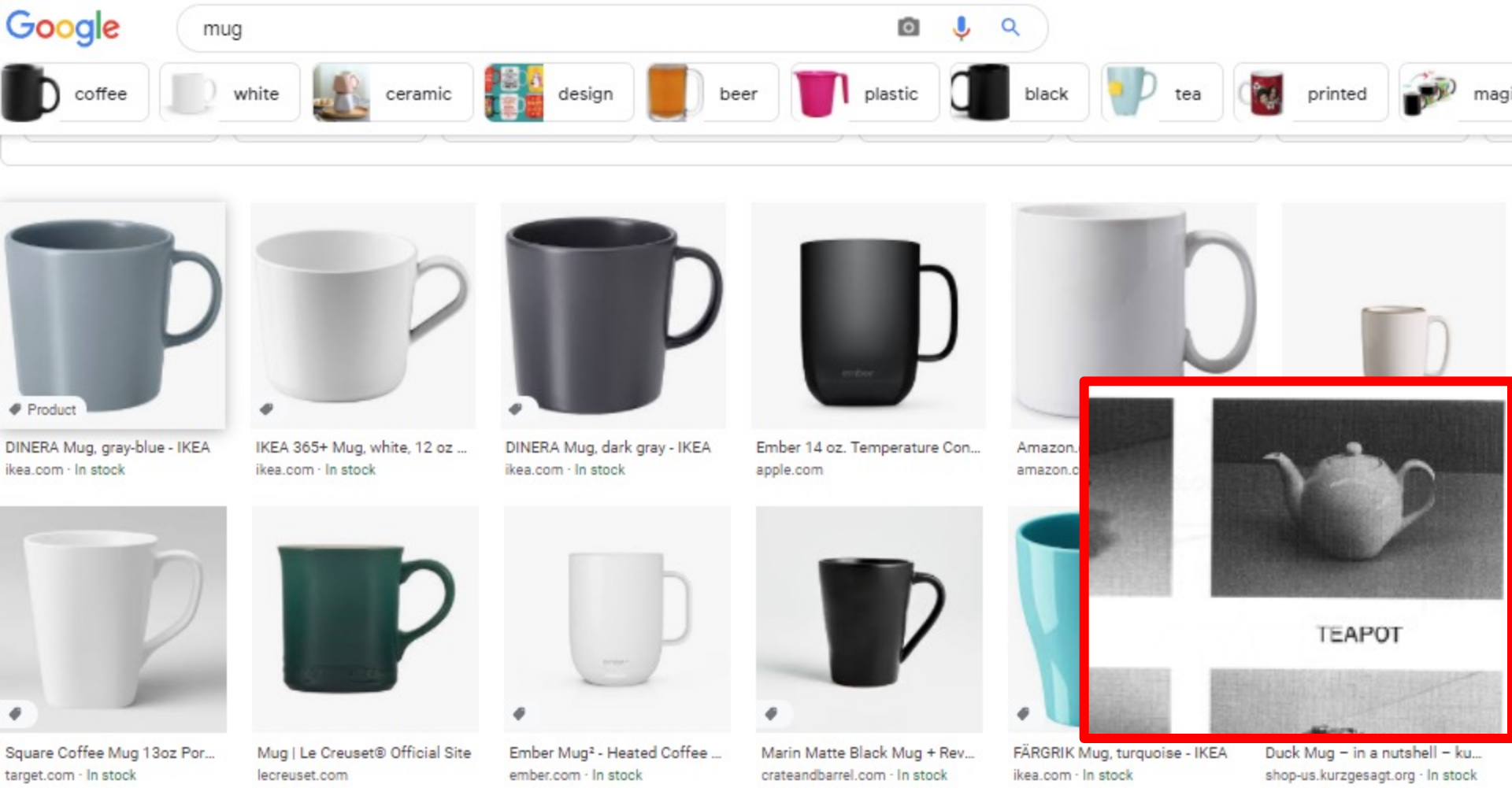


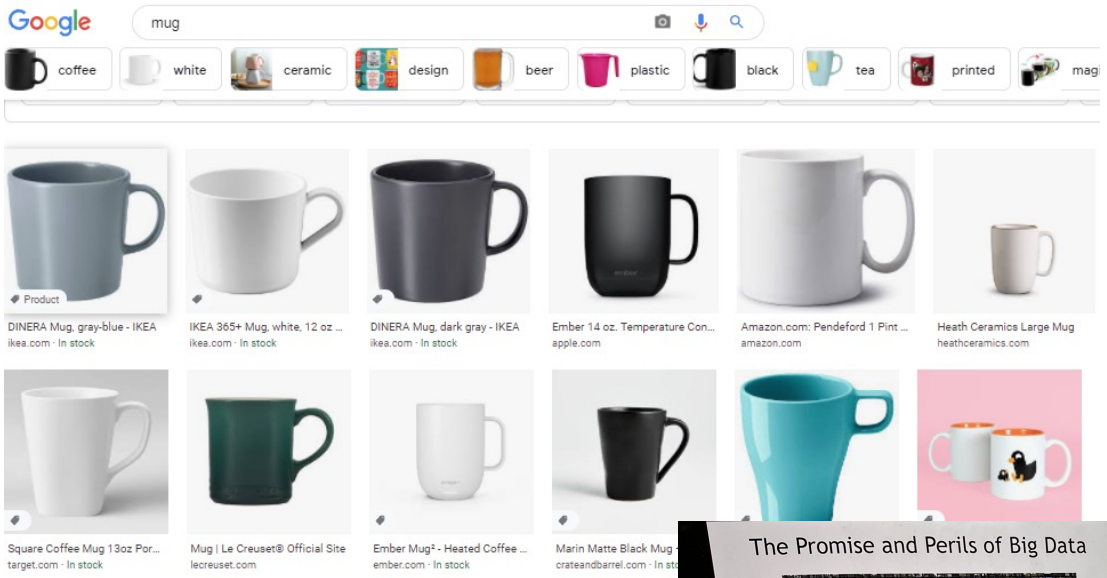
Salakhutdinov, Torralba, and Tenenbaum, CVPR, 2011

Some bias comes from the way the data is collected



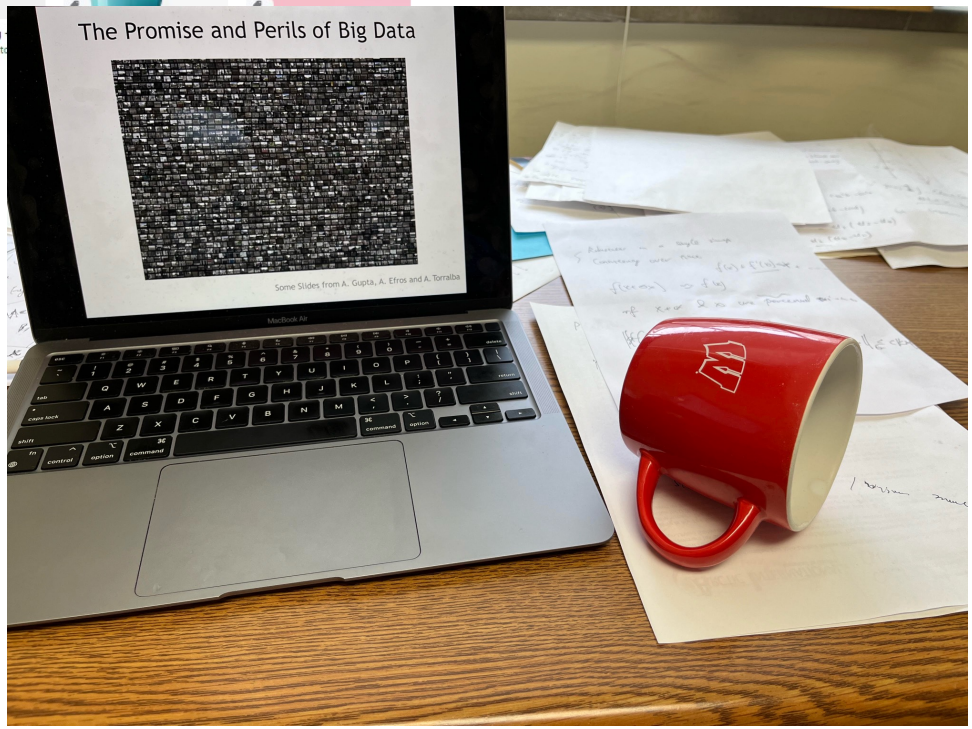
Some bias comes from the way the data is collected



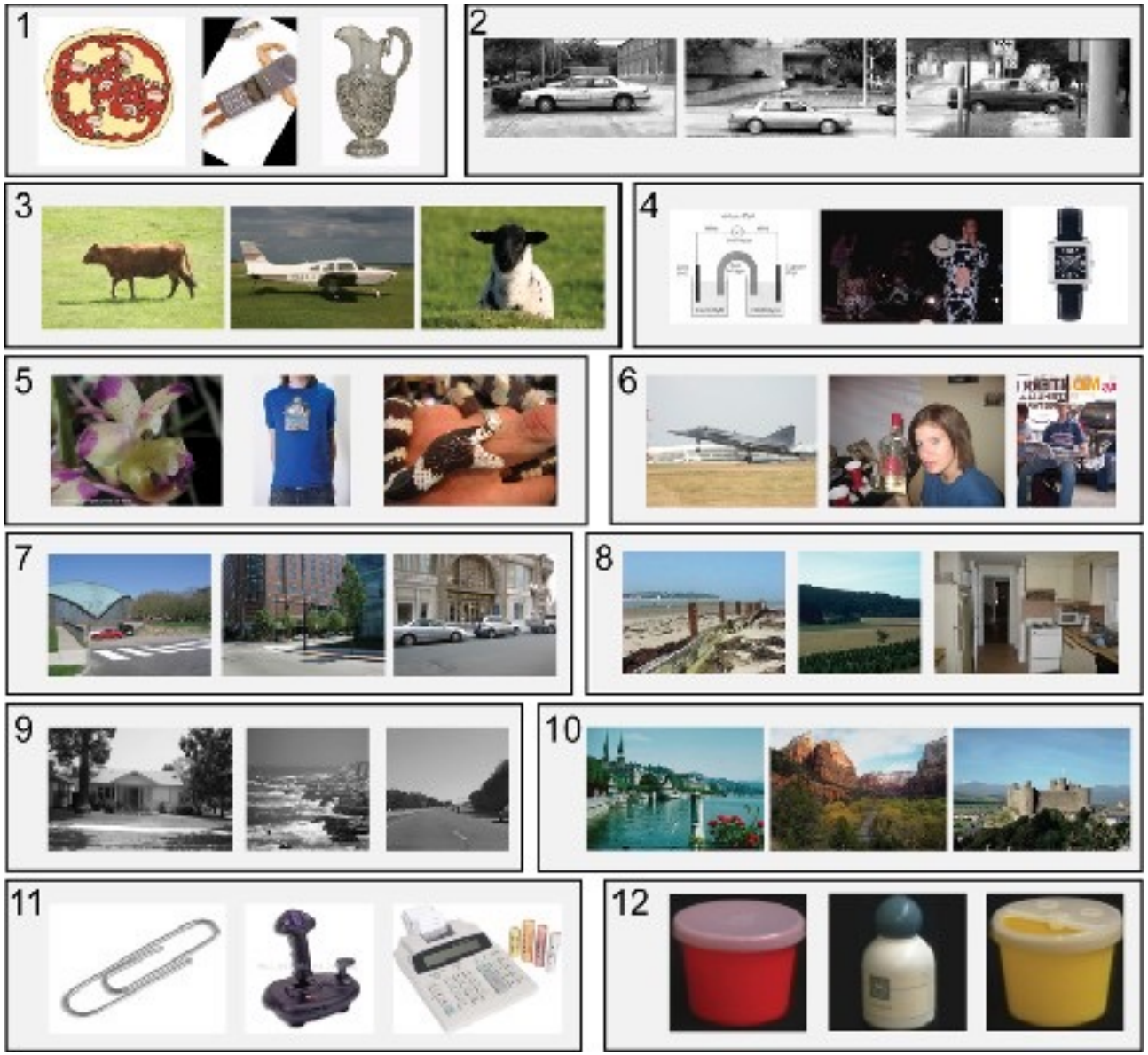


Mugs from Google

A mug in my office

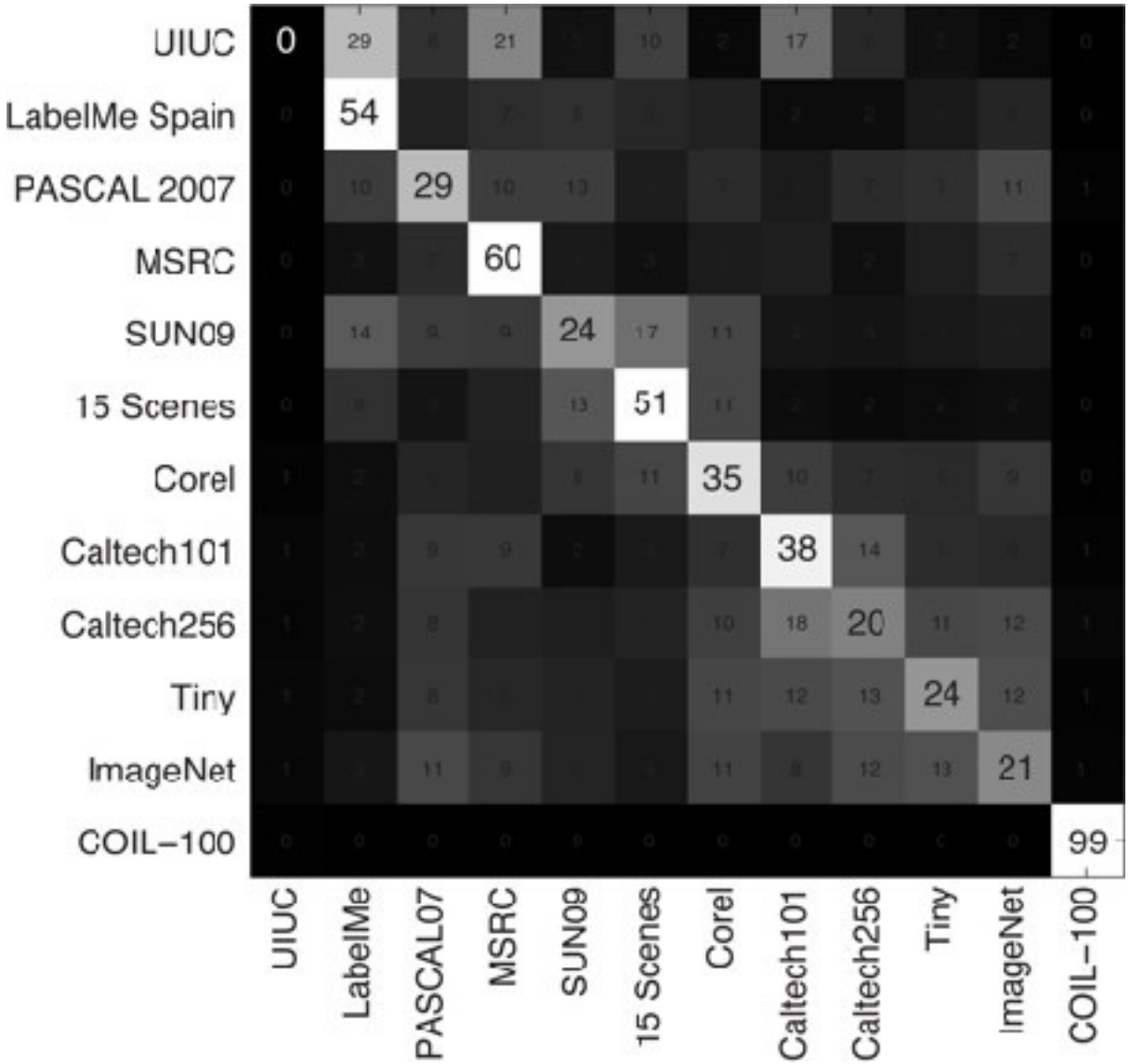


“*Name That Dataset!*” game



__ Caltech 101
__ Caltech 256
__ MSRC
__ UIUC cars
__ Tiny Images
__ Corel
__ PASCAL 2007
__ LabelMe
__ COIL-100
__ ImageNet
__ 15 Scenes
__ SUN’09

SVM plays *"Name that dataset!"*

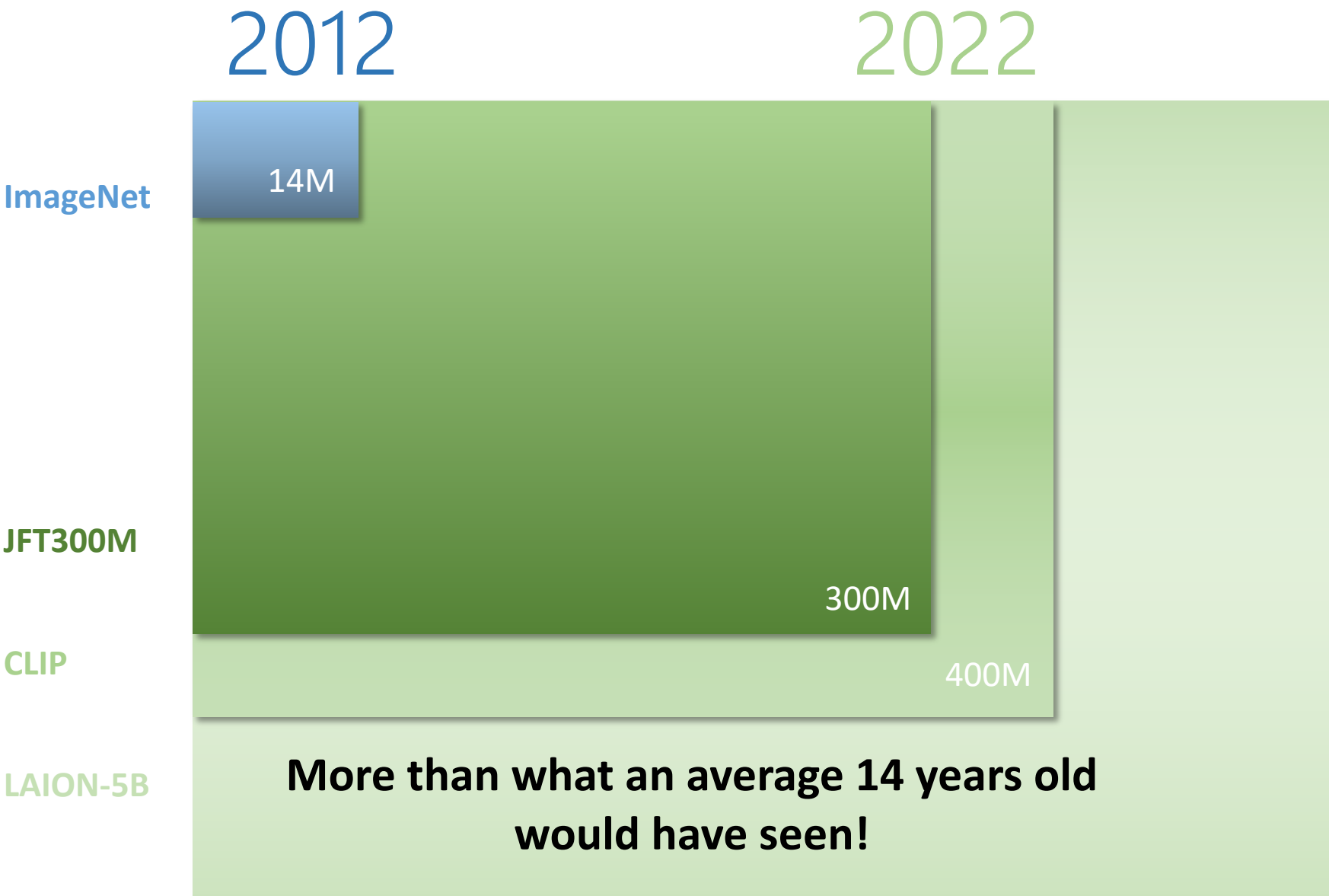


- 12 1-vs-all classifiers
- Standard full-image features

- 39% performance (chance is 8%)

Diminishing Returns of Data?

Images in vision datasets



Scaling up training data in Vision

More data leads to better generalizable representations

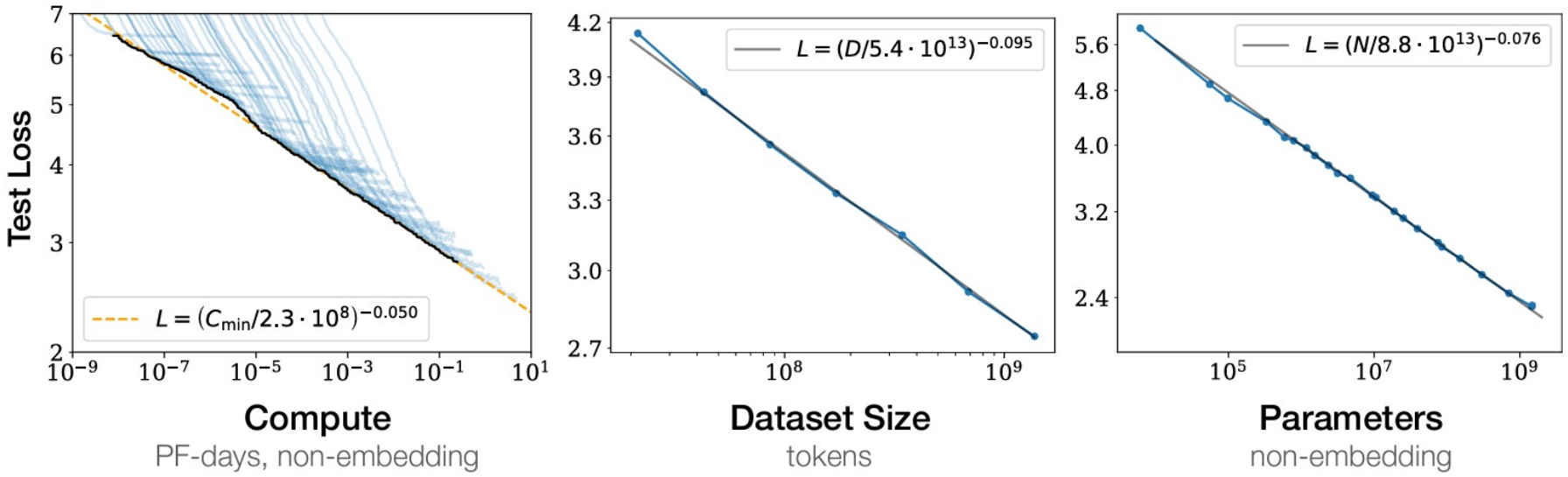
Vision task performance increase logarithmically based on volume of training data (ImageNet 14M -> JFT-300M)

Training w. long-tail distribution is mostly fine

But one has to grow the model size

Sun, C., Shrivastava, A., Singh, S., & Gupta, A. (2017). Revisiting unreasonable effectiveness of data in deep learning era. In 2017 IEEE international conference on computer vision (ICCV) (pp. 843-852). IEEE.

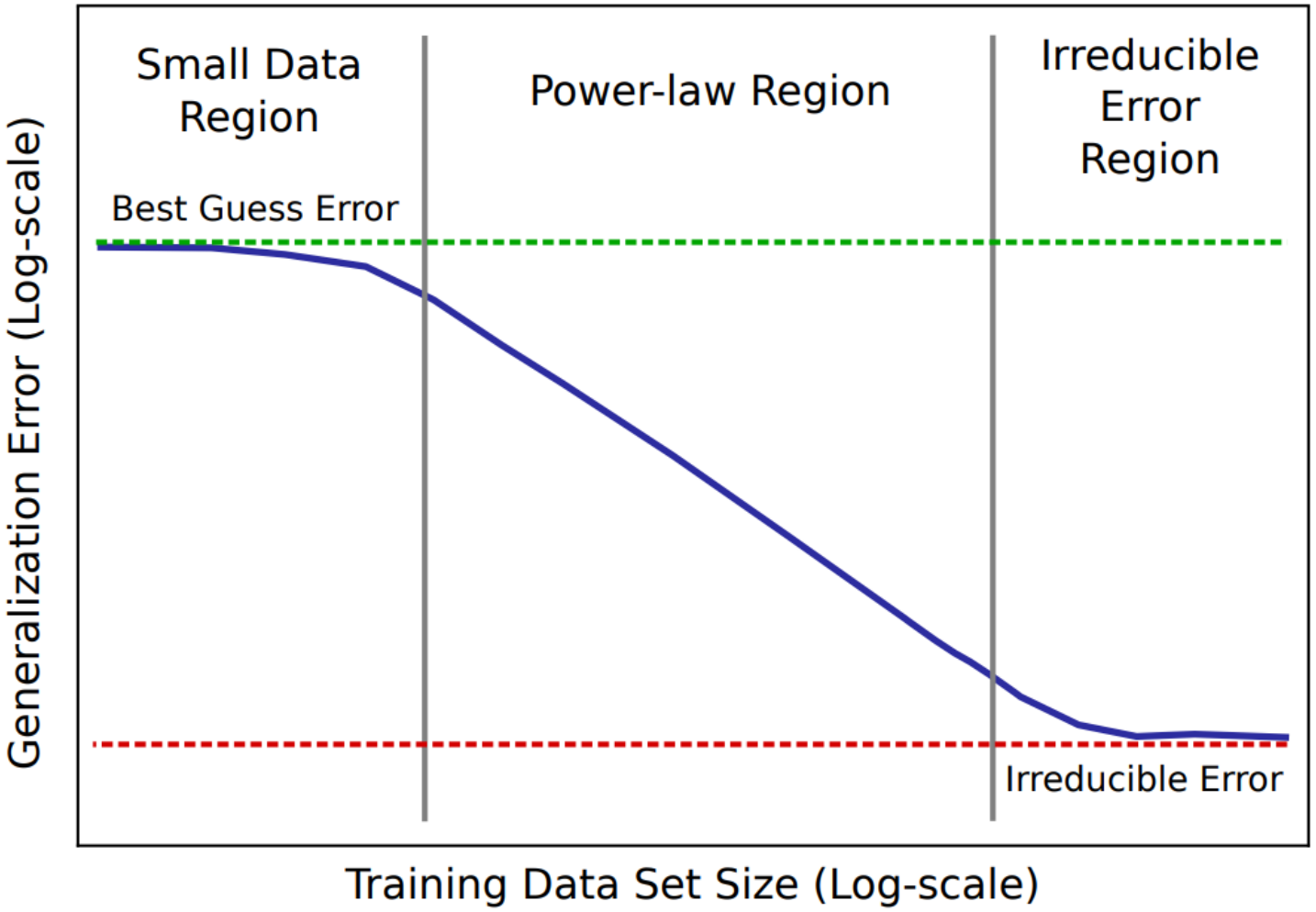
Scaling Laws (in NLP)



Language modeling performance improves smoothly as we increase the model size, dataset set size, and amount of compute used for training. For optimal performance, all three factors must be scaled up in tandem.

Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., ... & Amodei, D. (2020). Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*.

The Power Law of Model Scaling



Hestness, J., Narang, S., Ardalani, N., Diamos, G., Jun, H., Kianinejad, H., ... & Zhou, Y. (2017). Deep learning scaling is predictable, empirically. *arXiv preprint arXiv:1712.00409*.

- Why do we need data?
- How much data do we need?
- When do we need big data?
- How do we use big data?
- Why should we be careful about data?

The end